An initial segment of an isomorphism of well-orderings is an isomorphism onto the corresponding initial segment

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the full align computation and the closing preservation sentence are verbatim in the span; a commented-out earlier phrasing of the first equality remains as a source comment and delivers no text.

Editorial note. The label wellordinitialsegment names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Initial-segment notation. The unit's Definition before Lemma (source lines 765-772): A_a = {x in A : x < a} and <_a its restriction. Fixes the segment notation the statement and the calculation display both use.

Lemma 0.1. *Let $\langle A, < \rangle$ and $\langle B, \lessdot \rangle$ be well-orderings. If $f \colon A \to B$ is an isomorphism and $a \in A$, then $f \restriction A_a : A_a \to B_{f(a)}$ is an isomorphism.*

Proof. Since f is an isomorphism:

$$\begin{aligned} \operatorname{im}(f) A_a &= \operatorname{im}(f)\{\, x \in A \mid x < a \,\} \\ &= \operatorname{im}(f)\{\, f^{-1}(y) \in A \mid f^{-1}(y) < a \,\} \\ &= \{\, y \in B \mid y \lessdot f(a) \,\} \\ &= B_{f(a)} \end{aligned}$$

And $f \restriction A_a$ preserves order because f does. □